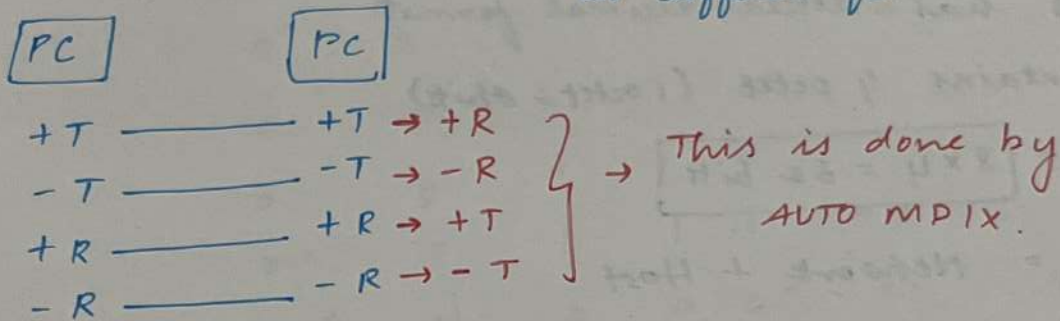


→ wired
 $\left\{ \begin{array}{l} \text{UTP} \\ \text{Fiber} \end{array} \right.$
 → Wireless

AUTO MDIX:

→ It can connect any cables into straight ^{or cross} even it is a same functional device; Eg: PC & Router or different functional device.



Day-3 IP & MAC

IP → Internet Protocol - Location of a device

MAC → Media access control - Identification of a device

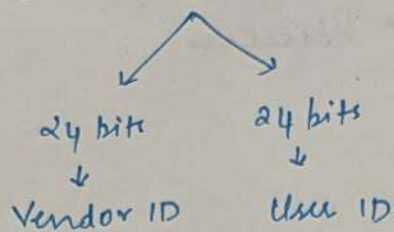
MAC:

- ↳ Physical address - Because it comes with the NIC card during manufacturing
- ↳ It is in hexadecimal format which has 4 bits & contains 12 hexadecimal.

So
 $12 \times 4 = 48 \text{ bits}$

Eg: 1C-C1-DE-33-8D-14 → 00011100 - 11000001 - 11011110 - 00110011 - 10001101 - 00010100

MAC has 48 bits



J. Bagavathi
↓ ↓
Vendor User

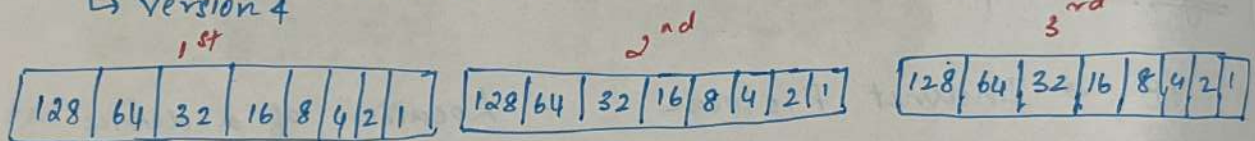
IP:

- ↳ It is a logical address → configured by us
- ↳ It has dotted decimal format
- ↳ Contains 4 octet (1 octet = 8 bits)

$$\Rightarrow 8 \times 4 = 32 \text{ bits}$$

IP = Network + Host

IPv4 structure:-



0.0.0.0 : Starting IP



255.255.255.255 : Ending IP

2-bit Binary:

2	1
---	---

0 0

0 1

1 0

1 1

3-bit Binary:

4	2	1
---	---	---

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

5-bit Binary:

16	8	4	2	1
----	---	---	---	---

0 0 0 0 0 → 0

0 0 0 0 1

0 0 0 1 0 → 2

0 0 0 1 1

0 0 1 0 0 → 4

0 0 1 0 1

0 0 1 1 0

0 0 1 1 1 → 8

0 1 0 0 0

0 1 0 0 1

0 1 0 1 0

0 1 0 1 1

0 1 1 0 0

0 1 1 0 1

0 1 1 1 0

0 1 1 1 1

1 0 0 0 0 → 16

1 0 0 0 1

1 0 0 1 0

1 0 0 1 1

1 0 1 0 0

1 0 1 0 1

1 0 1 1 0

1 0 1 1 1 → 23

6	8	4	2	1
---	---	---	---	---

2	1
---	---

Different classes of IP.

CLASS	RANGE	PORTION
A	0-127	N · H · H · H
B	128-191	N · N · H · H
C	192-223	N · N · N · H
D	224-239	Multicast
E	240-255	Broadcast R & D

} Invalid I.P

Eg:

- 1) 72.0.0.86 - A - N · H · H · H
- 2) 172.10.234.254 - B - N · N · H · H
- 3) 192.10.192.51 - C - N · N · N · H

50.0.0.10
 ↓ ↓
 N/W Host

8 8 8 8
 192.192.192.192
 N · N · N · H

Network - Street No.

Host - Door No.

Types of communication:

Unicast : One to One

Multicast : One to group

Broadcast : One to all

No. of network bits : 24

No. of host bits : 8

No. of network octet : 3

No. of host octet : 1